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PALEOMAGNETISM OF THE NEWARK TREND IGNEOUS ROCKS OF THE NORTH CENTRAL APPALACHIANS AND THE OPENING OF THE CENTRAL ATLANTIC OCEAN

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ABSTRACT. Paleomagnetic results from 28 sites on the Early Mesozoic Barndoor, East Rock, Mt. Carmel, and West Rock intrusives of the Hartford Basin, central Connecticut, yield VGPs in an area bounded by 52.1° to 116.8°E, 57.6° to 80.0°N. These results, along with previously published paleomagnetic data from Connecticut, suggest two periods of igneous activity, dated by radiometric and biostratigraphic methods at about 190 and 175 m.y. Paleomagnetic pole positions and ages of Newark trend igneous rocks elsewhere in the north central Appalachians suggest that the two igneous pulses were of regional extent, represented by mean pole positions (including the Connecticut results) located at 83.2°E, 63.0°N (190 m.y. age group) and at 103.2°E, 65.3°N (175 m.y. age group). These pole positions show about 9° of eastward apparent polar wander, in good agreement with results from paleomagnetic studies of Late Triassic-Middle Jurassic rocks in the western United States.

Rotation of Africa against North America in a pre-rift configuration produces an angular distance of only 2.5° between the mean pole positions of the younger (175 m.y.) group and a series of diabase dikes and sills in Liberia and Morocco dated at about 180 to 185 m.y., suggesting that active spreading of the Central Atlantic Ocean had not begun by 175 m.y. This is consistent with the early rifting nature of the Newark trend igneous suite and with interpretations based on borehole and sea floor magnetic anomaly data and regional structure which suggest that steady spreading probably began within the interval 170 to 175 m.y.

INTRODUCTION

The Early Mesozoic Barndoor, East Rock, Mt. Carmel, and West Rock diabase intrusives outcrop in the Hartford Basin of central Connecticut (figs. 1 and 2) and are probably related to the same major tectonic events that led to intrusion of Early Mesozoic Newark trend diabase dikes and sills elsewhere in the Appalachian piedmont and effusion of the basalt flows in the Newark Supergroup.

Prior paleomagnetic work on the Latest Triassic(?)-Jurassic diabase intrusives and basalt flows of Connecticut has been done by DuBois and others (1957), Bowker (ms), deBoer (1967, 1968), and Smith (1976). The first two were largely reconnaissance surveys in which little detailed demagnetization was carried out. deBoer (1967) studied diabase dikes along the length of the Appalachians, but his samples were subject to excessive AF demagnetization (deBoer, 1975, personal commun.), leading to obliteration of most of the original remanence. deBoer (1968) extensively studied the Meriden Formation basalt flows (Talcott, Hampden, and Holyoke basalts) of central Connecticut. Cornet (ms) shows that these units are of Hettangian-Sinemurian (Early Jurassic) age. In addition, deBoer (1968) did comparatively minor work on intrusives of the Hartford Basin, including the Mt. Carmel and West Rock bodies. He suggested that these units were emplaced before rotation of the host rocks

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to their present 20° to 30° eastward dip. Smith (1976) showed that the Connecticut narrow dikes (Bridgeport, Buttress, Cross Rock, Fairhaven, and Higganum dikes, fig. 2) were emplaced after most of the basin rocks were tilted. Sutter and Smith (this issue) calculated a mean age of 175 m.y. from these dikes, corresponding to Toarcian-Bajocian age. Cornet (ms) has suggested that portions of the Portland Formation (the youngest unit in the basin) were still being deposited at this time but that probably all deformation of the host rocks where applicable preceded dike intrusion.

The paleomagnetism of the Barndoor, East Rock, Mt. Carmel, and West Rock bodies was studied for several reasons:

1. To add to the body of paleomagnetic data already obtained from Newark trend igneous rocks of Connecticut with particular regard to augmenting the sparse data obtained from the Mt. Carmel and West Rock units by deBoer (1968).
2. To combine the new data with that already obtained from intrusive and extrusive basic rocks in the Newark trend (including results from Irving and Banks, 1961; Opdyke, 1961; Larochelle and Wanless, 1966; Beck, 1972; Carmichael and Palmer, 1968; in addition to those mentioned above) to determine if any distinct groups of VGP positions exist. The possibility of distinct groups arises from deBoer's (1968) suggestion that the West Rock and Mt. Carmel units were intruded before rotation of the basin rocks and Smith's (1976) conclusion that the narrow dikes were intruded following tilting of these rocks.
3. To determine whether paleomagnetic data from the Newark trend igneous rocks, along with recent sea floor magnetic data (Barrett and Keen, 1976), borehole data from offshore Nova Scotia (Given, 1977), and radiometric age data (Sutter and Smith, this issue) can be used to establish an earliest age of initial steady spreading of the Central Atlantic ocean, Dalrymple, Grommé, and White (1975) used paleomagnetic and radiometric age data from Newark trend rocks and rocks of similar age in Liberia and Morocco to estimate an initial opening age of 180 ± 10 m.y.

GEOLOGIC SETTING OF THE BARNDOOR EAST ROCK, MT. CARMEL, AND WEST ROCK UNITS

The intrusives were emplaced in the lower portion of the Norian-Rhaetian (Late Triassic) New Haven Formation of the Hartford Basin (fig. 2). They crop out in the western side of the basin and portions of the West Rock and Barndoor bodies contact Paleozoic metamorphic rocks forming the west edge of the basin. The Barndoor intrusive has an outcrop length of 35 km and attains a maximum thickness of about 300 m; the West Rock intrusive has a similar maximum thickness and an outcrop length of about 30 km. Field relationships and aeromagnetic data (U.S. Geol. Survey, 1966, 1973) suggest that they are largely conformable with the host rocks which dip 20° to 30° east-southeast. Both

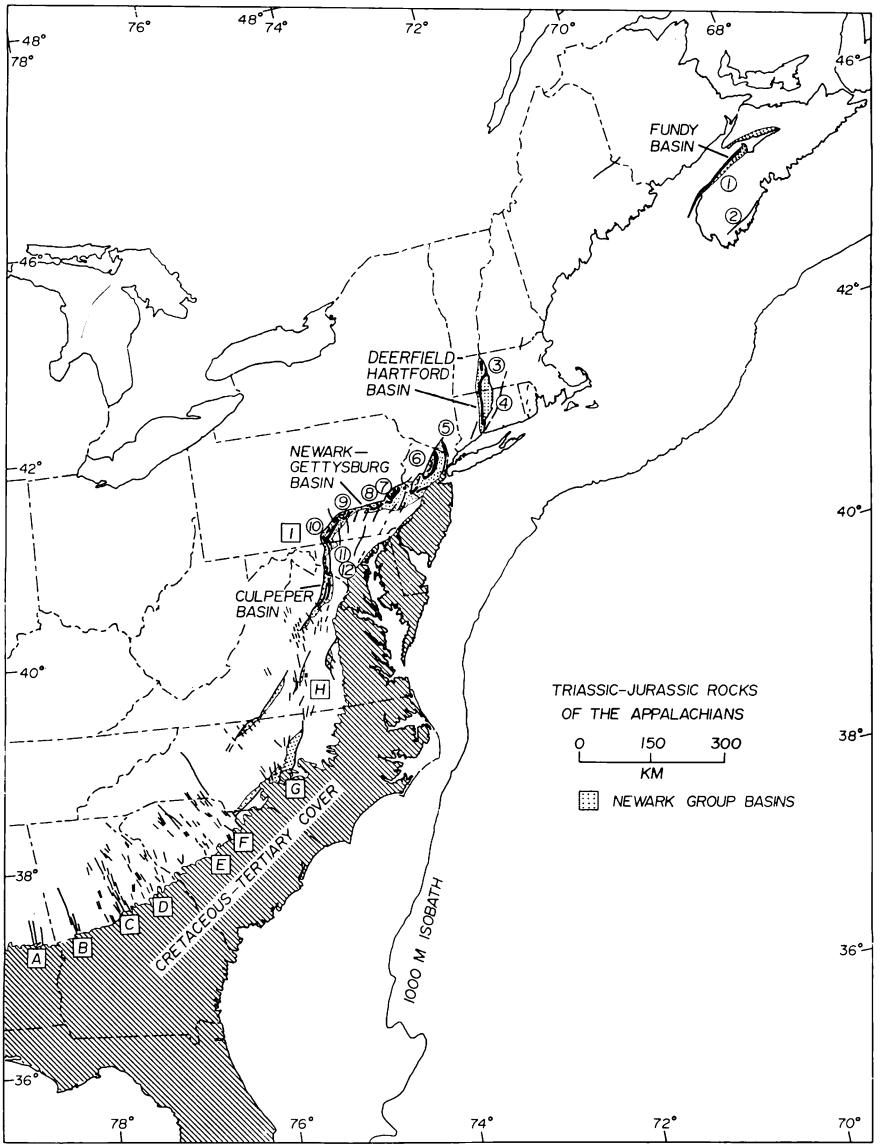


Fig. 1. Outline geologic map of the Triassic-Jurassic rocks of the Appalachians. Igneous units: 1 — North Mountain basalts. 2 — Shelburne dike. 3 — Holyoke, Deerfield, and Granby volcanic units. 4 — Holyoke, Talcott, and Hampden volcanic units; East Rock, Mt. Carmel, and West Rock intrusives; Barndoor, Bridgeport, Buttress, Cross Rock, Fairhaven, and Higganum dikes ("narrow dikes"). 5 — Palisades sill. 6 — Watchung basalts. 7 — Quakertown sill. 8 — Birdsboro sill. 9 — York Haven sill. 10 — Gettysburg sill. 11 — Frederick dikes. 12 — Boyds sill. A-I: dike swarms classified by King (1971).

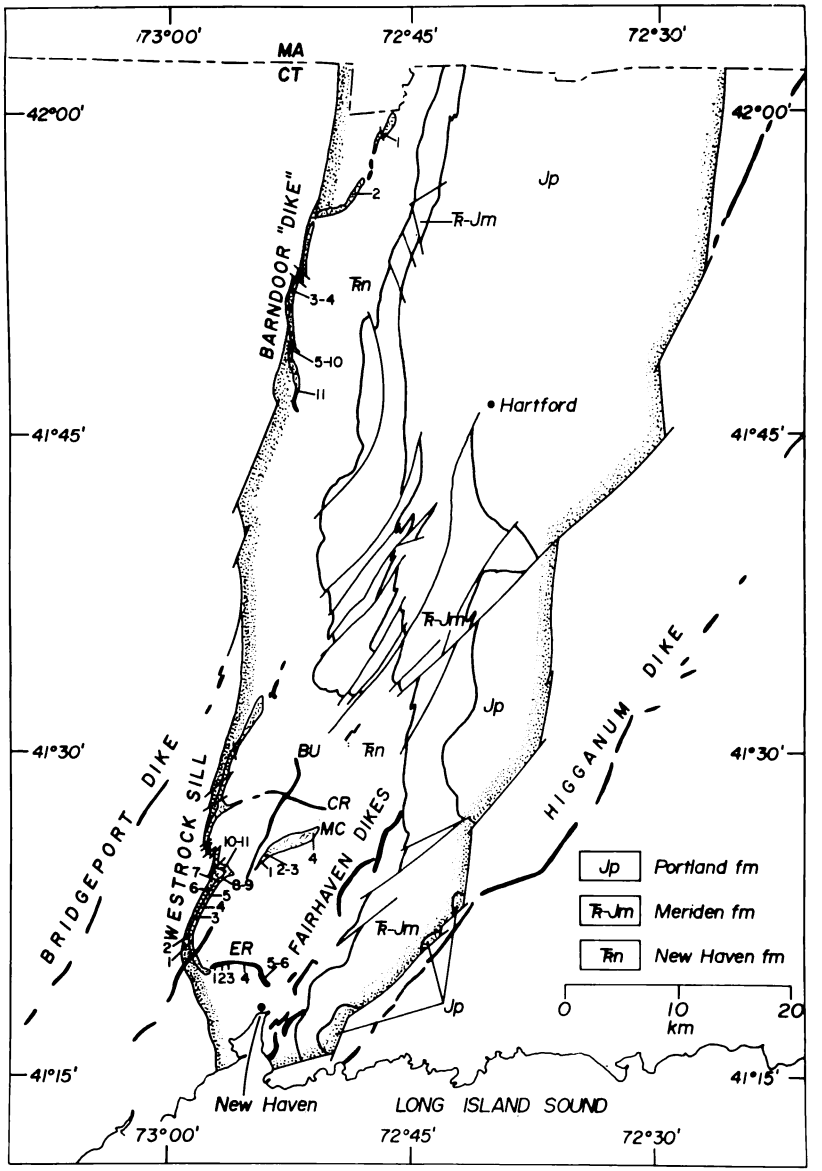


Fig. 2. Diabase intrusives of the Hartford Basin and sampling sites. Numbers correspond to those given in table 1. The edges of the Hartford Basin are stippled.

units form prominent topographic features and are cut by numerous small northwest-southeast trending faults, parallel to faults of large displacement cutting the Meriden Formation basalt flows which overlie the New Haven Formation. Sandstone dikes and steam blow holes present at the lower contact of the West Rock body suggest intrusion in wet, soft sediments (Rodgers, 1975, personal commun.). The West Rock intrusive is cut by the Buttress dike, a member of the narrow dike group.

The East Rock body crops out 3 km north of New Haven (fig. 2) and has a discontinuous east-west outcrop of about 6 km. It is well exposed along most of its length; northwest of New Haven it consists of a complicated anastomosing network and is extensively quarried. The Mt. Carmel body crops out about 16 km north of New Haven over an area of about 4 km^2, contained mostly in Sleeping Giant State Park. Fritts (1963) suggested that it is a dome-shaped portion of the West Rock sheet, whose main outcrop is about 3 km west. Aeromagnetic data (U.S. Geol. Survey, 1966, 1973) suggest that both the Mt. Carmel and East Rock bodies are portions of the West Rock sheet.

ANALYTICAL TECHNIQUES

Between 4 and 15 25-mm diam cores were drilled at each of 33 sites on the 4 intrusives (fig. 2), yielding a total of 221 cores and 301 25 × 23 mm specimens. The number of specimens thus obtained from each site ranged from 5 to 15. Sampling at each site covered 3 to 10 m, and recent road cuts were sampled where possible. Specimens were measured by a Schonstedt SSM-1A5 Hz magnetometer and demagnetized in a Schonstedt GSD-1 AF demagnetizer. Following measurement of NRM directions, a pilot suite of 5 specimens from each site was demagnetized in 50 œ steps from 50 to 250 œ, and that treatment producing the highest value of k consistent with paleomagnetic stability was used to clean all specimens from the site. Structural corrections were made by rotating specimen magnetic directions by an amount corresponding to the dip of the host rocks. This simple correction is sufficient as most of the basin rocks show little or no folding around sub-vertical axes.

Additional preparation and analysis were carried out on representative specimens to obtain further insight on magnetic characteristics and mineralogy. Eight specimens were demagnetized in 100 œ steps from 300 to 1000 œ. These specimens had been previously demagnetized in 50 œ steps to 200 œ as members of pilot suites of various sites. Another group of 8 specimens was thermally demagnetized (ambient field of 10 γ) to 400°C in 100° steps, thereafter to 600°C in 50°C steps. Six polished sections, two each from the Barndoor and West Rock intrusives, and one each from the East Rock and Mt. Carmel intrusives, were examined with a Reichert Metapan ore microscope at 900 magnification with oil immersion. A chip from each section was mapped for Fe and Ti in a Cambridge Stereoscan 54-10 SEM, using an Ortec multichannel analyzer to obtain relative elemental abundances. All analytical work was done at The Ohio State University.

The following criteria were used for rejecting results:

1. Clustering of magnetic directions about 2 or more separate means, or about the present field directions, following cleaning.
2. Instability persisting up to 200 œ.
3. High NRM moments ($>7.5 \times 10^{-2}$ gauss) which may be indicative of lightning strikes.
4. Site α-95 radii exceeding 10° after optimum cleaning.

These criteria are very similar to those used by Beck (1972) and Smith (1976) in extensive paleomagnetic studies of Newark trend intrusives of Connecticut, Pennsylvania, and Maryland, ensuring uniformity in the quality of the data.

DEMAGNETIZATION SPECTRA AND MAGNETIC MINERALOGY

AF demagnetization spectra of 8 representative specimens (fig. 3) show an average normalized residual intensity of 71 percent following cleaning in 50 œ, 22 percent following 150 œ, and 1 percent following 1000 œ. The AF response of these specimens typically resembles that of the Rossville diabase type of Pennsylvania (Volk, 1977), considered to be typical of primarily multi-domain and pseudo-single domain behavior. A few of the samples showed spiky demagnetization curves, a possible result of selective removal of oppositely directed components during demagnetization. The generally low coercivity of most of the remanence is typical of many of the Newark trend intrusives (deBoer, 1968; Beck, 1972; Watts, ms; Smith, 1976) and suggests that much of the NRM consists of overprinting by the present field.

Thermal demagnetization spectra (fig. 3) revealed very consistent behavior with blocking temperatures in the range of 500° to 600°C and an average residual normalized intensity of 74 percent following heating to 400°C. Thereafter, intensities declined rapidly, and the specimens became dark brown due to oxidation of iron bearing minerals. Three of the specimens showed only minor loss of intensity by 450°C, followed by a sharp decrease above this temperature. This behavior may be an indication of thermally induced breakdown of titanomaghemite to magnetite and ilmenite in the range of 350° to 450°C (Johnson and Merrill, 1973; Stacey and Bannerjee, 1974). The high blocking temperatures indicate alteration of the original magnetic composition, probably titanomagnetite similar to that in fresh sea floor basalts, by exsolution and oxidation during cooling.

Opaque microscopy of the 6 polished sections showed that 60 to 80 percent of the iron-titanium oxide grains contained patchy intergrowths of titanomaghemite and medium to broad ilmenite lamellae, corresponding to classes I and II of Wilson and Watkins (1967). These classes are indicative of only slight high temperature (1100°-500°C) oxidation. SEM microscopy of 6 chips from these specimens revealed large regions (up to 200 μ) of relatively pure magnetite, probably a residuum following exsolution of ilmenite from the original titanomagnetite composition. This magnetite, that produced by thermal breakdown of titanomaghemite,

and perhaps some single domain (?) magnetite incorporated in plagioclase (Hargraves and Young, 1969) is responsible for the observed high blocking temperatures. These results indicate that much of the diabase remanence is probably CRM acquired during slow cooling and accompanying oxidation, resulting in reduction of short term magnetic (secular variation) scatter.

PALEOMAGNETIC RESULTS

NRM directions from most sites confirmed moderate to considerable remagnetization in the present field. NRM intensities ranged between

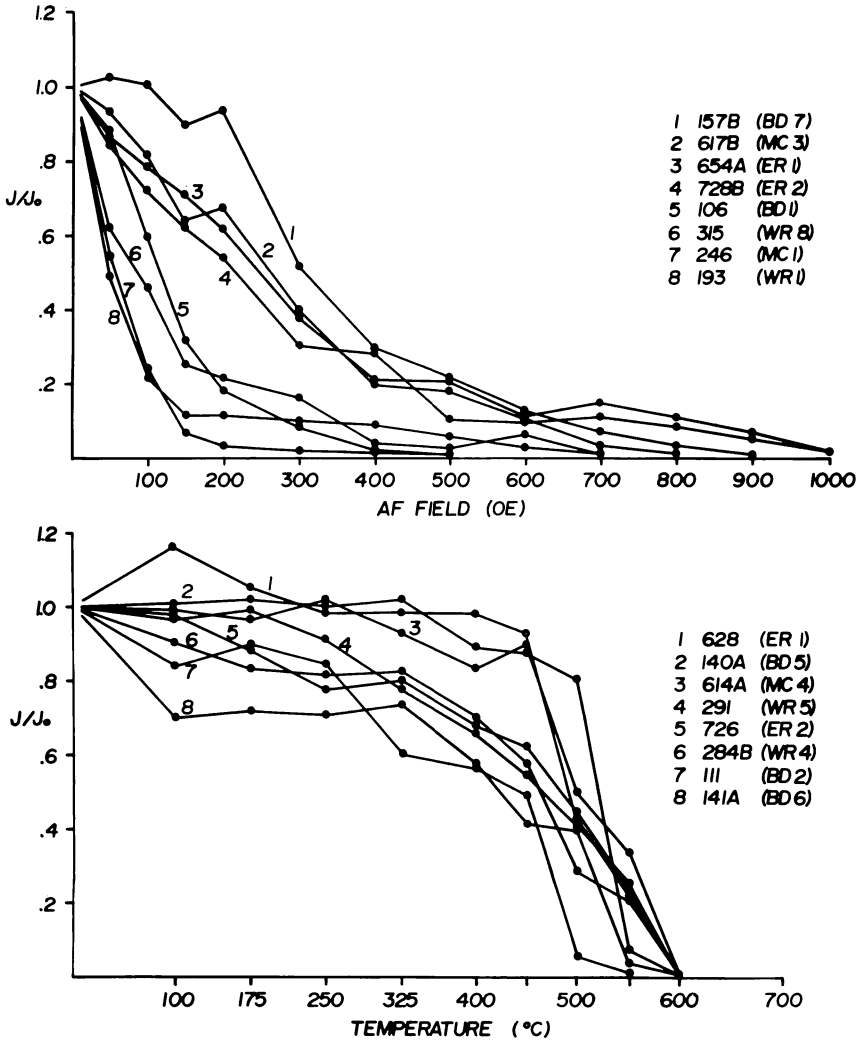


Fig. 3. AF and thermal decay curves for representative specimens. Specimens and site number are given for each curve. Both sets of curves are normalized to NRM intensity.

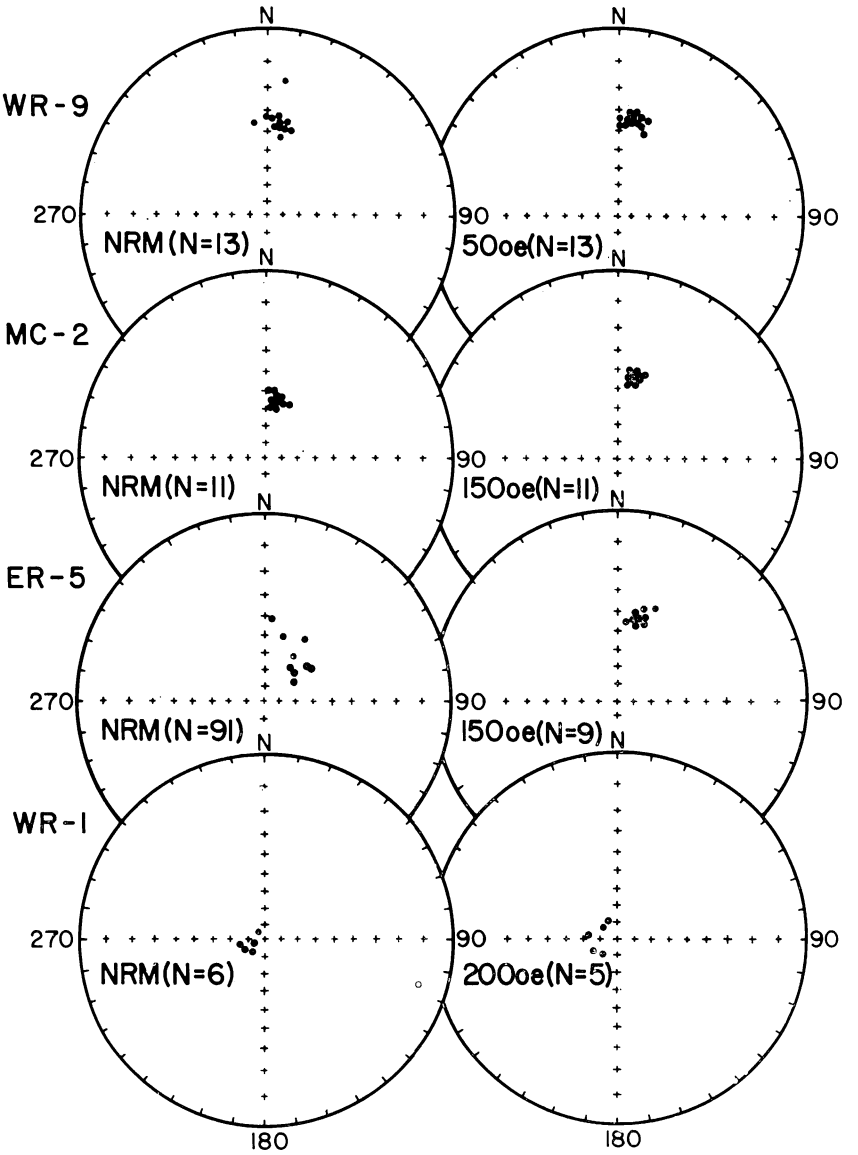
3.5×10^{-1} and 1.4×10^{-3} gauss. Results from only 1 site (BD-2) suggested reversed overprinting. Most NRM directions clustered well (figs. 4 and 5) with NRM α-95 less than 10° for 25 of the 33 sites. Mean NRM field directions of these sites (fig. 5) often lay between those of the present field at the sampling localities (D = 345°, I = 70°) and those of Late Triassic-Middle Jurassic time (D = 350–10°, I = 20–40°).

Following AF demagnetization, the paleomagnetic results of 5 sites were rejected by the criteria given above. Results from 5 additional sites were accepted after rejection of 1 to 4 specimens displaying erratic behavior. The directions of these specimens were more than 10° away from respective site mean directions and may have been the result of specimen misidentification or misorientation.

Nearly all optimum cleaning treatments ranged between 50 and 150 œ; specimens from only 1 site (WR-12) required cleaning in 200 œ. Cleaning in stronger fields produced only slight changes in site mean directions but with increasing scatter. In most cases, the AF demagnetization resulted in less than 15° of motion of the site mean directions from their NRM positions, with most motion usually occurring by cleaning in 50 œ. All but 1 of the accepted sites had α-95 radii less than 8°, 8 had α-95 radii of 3° or less, and all were of normal polarity.

Site paleomagnetic data and VGPs of all intrusives for both uncorrected and structurally corrected situations are listed in table 1; VGPs obtained from reliable results are shown in figure 5. The VGPs from the East Rock, Mt. Carmel, and West Rock intrusives cluster well, particularly so following structural correction. Previous results from the Mt. Carmel and West Rock bodies (app 1) obtained by deBoer (1968) are in good agreement. The mean pole positions of all 3 units (including deBoer's 1968 data), when corrected for structure, are located in the vicinity of 70° to 85° E, 63° to 67°N, within 7.7° of each other (table 2 and fig. 6) and are statistically indistinguishable at the 95 percent level of confidence. The mean pole position of the structurally corrected VGPs lies at 82.5°E, 63.1°N.

In the absence of structural correction, the mean pole positions of these units are located in the vicinity of 96° to 109°E, 50° to 65°N and are separated by up to 15.4° (table 2, fig. 6). The paleomagnetic results suggest, therefore, that the East Rock, Mt. Carmel, and West Rock bodies were emplaced before rotation of the host rocks, consistent with previously mentioned aeromagnetic evidence that they are exposures of a single large intrusive sheet and field evidence showing emplacement of the West Rock body in wet sediments. In addition, the structurally corrected VGPs of these units are very similar to those from the Talcott, Hampden, and Holyoke basalts (deBoer, 1968) listed in appendix 1. The flows currently dip 20° to 30°E but were probably extruded on a nearly level surface. Radiometric age determinations (both conventional K–Ar and $^{40}Ar/^{39}Ar$) on both the West Rock and Mt. Carmel units listed in Sutter and Smith (this volume) agree well with the Hettangian-Sinemurian age of the flows. Addition of these results from the flows to those


WR-9
NRM(N=13)
50oe(N=13)
MC-2
NRM(N=11)
150oe(N=11)
ER-5
NRM(N=91)
150oe(N=9)
WR-1
NRM(N=6)
200oe(N=5)
N
90
270
180

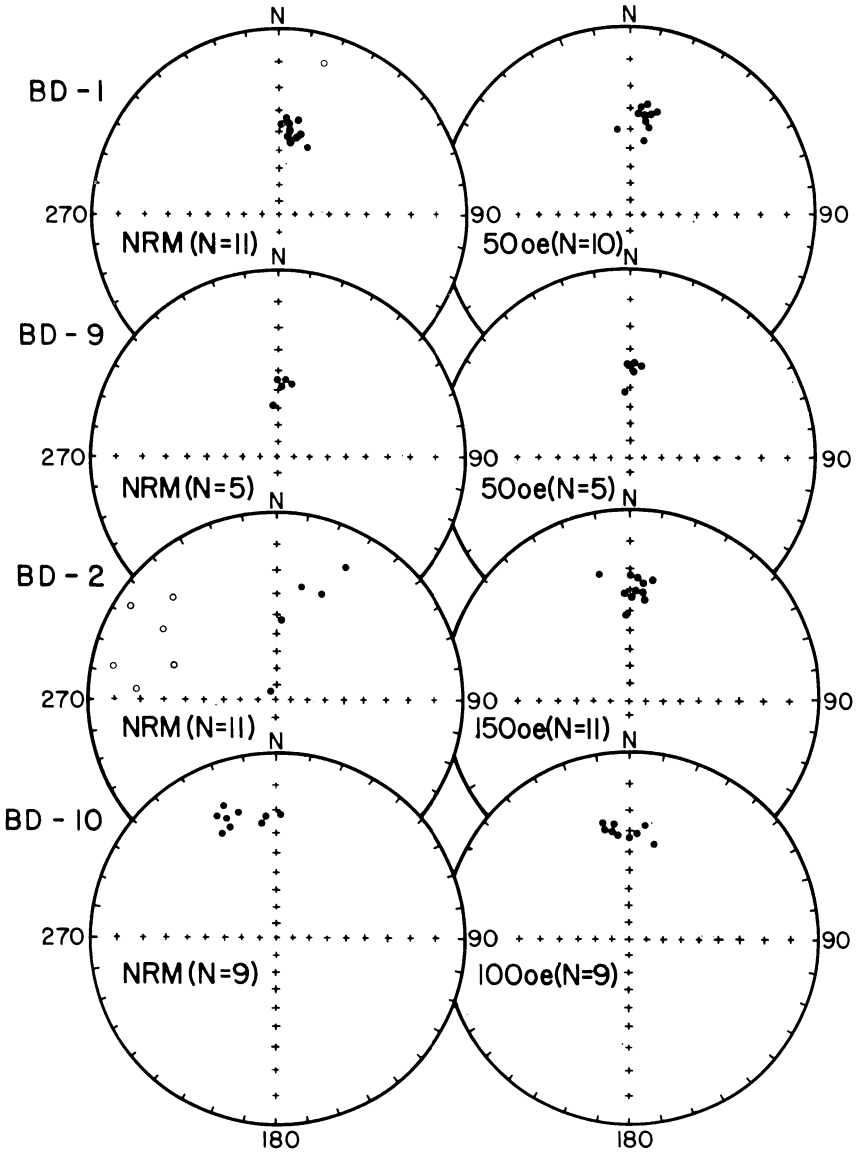


Fig. 4. NRM and cleaned directions for 8 representative sites (site numbers correspond to those in table 1). Note the generally good clustering of the NRM directions and absence of negative inclinations (open circles) in the cleaned results. Polar stereographic projection.

TABLE 1

Site paleomagnetic data from the West Rock, East Rock, Mt. Carmel, and Barndoor intrusives of Connecticut

							no structural correction				structural correction					
									VGP position				VGP position			
Site	Long	Lat	N	H	k	α-95	Dec	Inc	Long	Lat	Dec	Inc	Long	Lat	Dipdir	Dip
West Rock intrusive																
WR-1	72°58'	41°20'	6	150	75	7.8	014.5	31.7	75.6	62.7	027.7	34.9	52.1	57.6	120°	20°
WR-2	72°58'	41°22'	8	150	354	2.9	354.0	19.1	117.7	56.4	001.8	29.8	103.1	62.9	120°	20°
WR-3	72°47'	41°22'	9	100	178	3.9	359.6	26.0	107.7	60.7	010.3	34.6	83.8	64.5	120°	20°
WR-4	72°57'	41°23'	10	150	131	4.2	358.6	27.0	109.8	61.3	009.8	35.8	84.4	65.4	120°	20°
WR-5#	72°57'	41°23'	5	150	10	25.6	007.3	58.7	39.8	83.9	041.3	59.0	8.0	59.0	120°	20°
WR-6#	72°56'	41°24'	12	200	9	15.5	025.3	10.5	69.3	46.9	028.9	9.8	65.1	44.8	120°	20°
WR-7	72°56'	41°24'	7	50	62	7.7	347.6	31.1	134.4	62.6	000.4	41.5	106.8	71.8	120°	20°
WR-8 (1)	72°56'	41°24'	8	100	806	1.9	356.4	31.8	116.2	65.0	009.7	41.0	81.4	69.8	120°	20°
WR-9	72°56'	41°24'	13	50	315	2.3	356.0	24.8	115.9	60.8	007.6	35.5	89.7	66.7	120°	20°
WR-10#	72°56'	41°24'	5	200	7	32.1	098.6	62.6	-21.7	22.6	107.6	38.6	-9.6	2.2	120°	20°
WR-11#	72°56'	41°24'	6	200	5	34.1	022.7	87.8	-69.6	46.0	114.8	65.2	-32.0	16.2	120°	20°
WR-12 (4)	72°54'	41°32'	9	200	134	4.5	344.9	38.4	144.6	66.0	006.5	52.9	74.7	80.0	120°	25°
East Rock intrusive																
ER-1	72°57'	41°21'	14	150	261	2.5	006.2	27.3	95.2	62.0	020.8	34.6	64.1	61.0	120°	25°
ER-2 (1)	72°57'	41°21'	12	150	205	3.0	354.7	28.0	119.2	62.5	009.2	43.7	80.4	71.9	130°	25°
ER-3	72°56'	41°21'	8	50	139	4.7	357.6	38.3	114.4	69.5	013.4	38.4	74.9	66.7	120°	25°
ER-4	72°56'	41°21'	9	150	226	3.4	358.0	28.3	112.2	63.0	017.4	39.6	66.0	65.6	120°	30°
ER-5	72°53'	41°20'	9	150	257	3.2	359.4	33.7	109.4	66.5	013.8	41.7	71.0	68.7	120°	20°
ER-6	72°53'	41°20'	9	150	284	3.1	359.1	34.5	110.2	66.9	013.9	42.6	70.1	69.2	120°	20°
Mt. Carmel intrusive																
MC-1#	72°54'	41°25'	6	150	6	29.1	316.1	39.3	-176.4	48.5	324.5	63.5	-142.9	64.3	125°	25°
MC-2	72°54'	41°25'	10	150	485	2.2	355.9	34.1	117.6	66.5	010.5	43.4	77.5	71.1	120°	20°
MC-3	72°54'	41°25'	9	150	126	4.6	004.6	24.5	98.6	59.6	017.5	30.7	72.7	59.6	115°	25°
MC-4	72°52'	41°26'	14	150	126	3.6	000.1	23.8	107.9	60.4	009.4	30.7	87.6	63.3	130°	15°
Barndoor intrusive																
BD-1 (1)	72°47'	41°58'	10	50	118	4.5	007.9	34.9	89.0	66.2	027.2	40.4	49.3	60.5	120°	25°
BD-2	72°48'	41°57'	11	150	101	4.6	003.2	28.9	101.5	63.3	011.7	32.2	82.5	63.5	110°	15°
BD-3 (4)	72°52'	41°52'	11	50	424	2.2	014.7	31.9	76.6	62.3	028.9	33.2	53.2	55.7	110°	20°
BD-4	72°52'	41°52'	6	150	59	7.4	011.5	22.4	85.3	58.0	020.1	33.9	70.4	55.7	110°	20°
BD-5	72°52'	41°49'	8	150	78	6.3	356.8	31.2	115.5	64.7	009.9	40.3	81.5	69.3	120°	20°
BD-6	72°52'	41°49'	9	150	29	9.7	006.1	32.9	93.9	65.4	020.1	38.8	61.7	63.8	120°	20°
BD-7	72°52'	41°49'	8	150	857	1.9	358.7	28.3	110.9	63.1	010.4	37.1	82.6	66.9	120°	20°
BD-8	72°52'	41°49'	8	50	56	7.4	010.6	37.7	81.6	67.3	027.7	41.6	48.2	61.2	120°	20°
BD-9	72°52'	41°49'	5	150	152	6.7	001.9	40.6	104.9	71.2	019.6	47.6	52.0	69.2	120°	20°
BD-10	72°52'	41°49'	9	100	77	5.9	355.9	29.6	116.8	63.6	008.1	39.1	86.6	68.9	120°	20°
BD-11	72°52'	41°47'	7	50	354	3.2	008.9	29.2	69.3	62.5	021.4	34.5	63.8	60.8	120°	20°

Results were discarded for reasons given in the text. N — Number of specimens. H — Cleaning field, oe. k — Fisherian precision parameter. α-95 — Radius of cone of 95 percent confidence. Dipdir — Dip direction. Numbers in parentheses give number of specimen results discarded following cleaning.

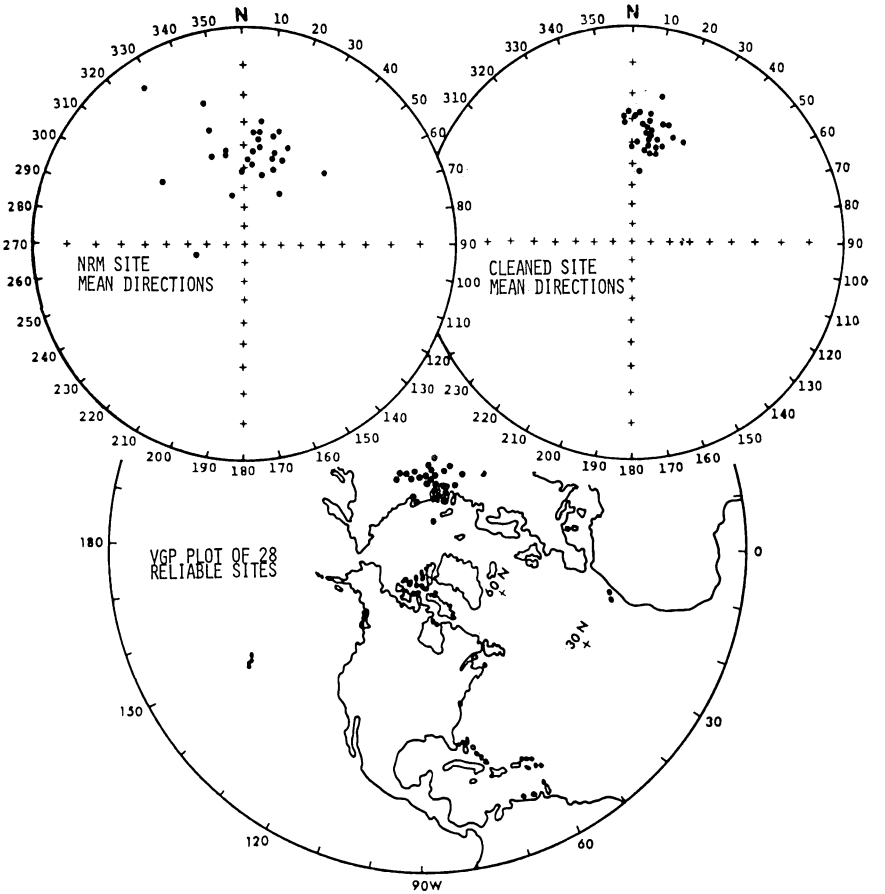


Fig. 5. Mean NRM and cleaned directions and plot of the site mean VGPs for the 28 reliable sites. All results have been corrected for structure except those from the Barndoor intrusive. All data are from table 1. Polar stereographic plots.

TABLE 2

Mean pole positions of the Barndoor, East Rock, Mt. Carmel and West Rock intrusives

		No structural correction				Structural correction			
Intrusive	N	k	α_{95}	E. Long	N. Lat	k	α_{95}	E. Long	N. Lat
Barndoor	11	146	3.8°	96.7°	64.9°	103	4.5°	64.7°	53.5°
Westrock	(8)11*	42	7.1°	109.4°	58.3°	86	4.9°	86.0°	63.2°
Mt. Carmel	(3) 8*	48	8.1°	103.6°	50.6°	123	5.0°	87.3°	66.7°
Eastrock	6	336	3.7°	109.7°	65.7°	345	3.6°	70.4°	67.3°

Mean pole positions calculated by averaging VGP's. N is number of site VGP's.
* Number in parentheses is the number of sites sampled in this study. The remaining sites were sampled by deBoer (1968).

of the East Rock, Mt. Carmel, and West Rock units produces a mean paleomagnetic pole position located at 83.9°E, 62.6°N, only 0.8° from that of the intrusives alone.

The results from the Barndoor intrusive are different from those of the other three bodies. Its structurally corrected mean pole position of 96.7°E, 64.9°N (table 2) is 15° or more away from those of the East Rock, Mt. Carmel, and West Rock bodies and is statistically distinguishable from them (fig. 6). However, its uncorrected mean paleomagnetic pole position located at 96.7°E, 64.9°N is indistinguishable from the mean paleomagnetic pole of the Bridgeport, Buttress, Cross Rock, Fairhaven, and Higganum dikes (narrow dikes) which is located at 99.7°E, 69.3°N (Smith, 1976). This suggests that the Barndoor intrusive was emplaced at the same time as the narrow dikes, following tilting of most of the basin

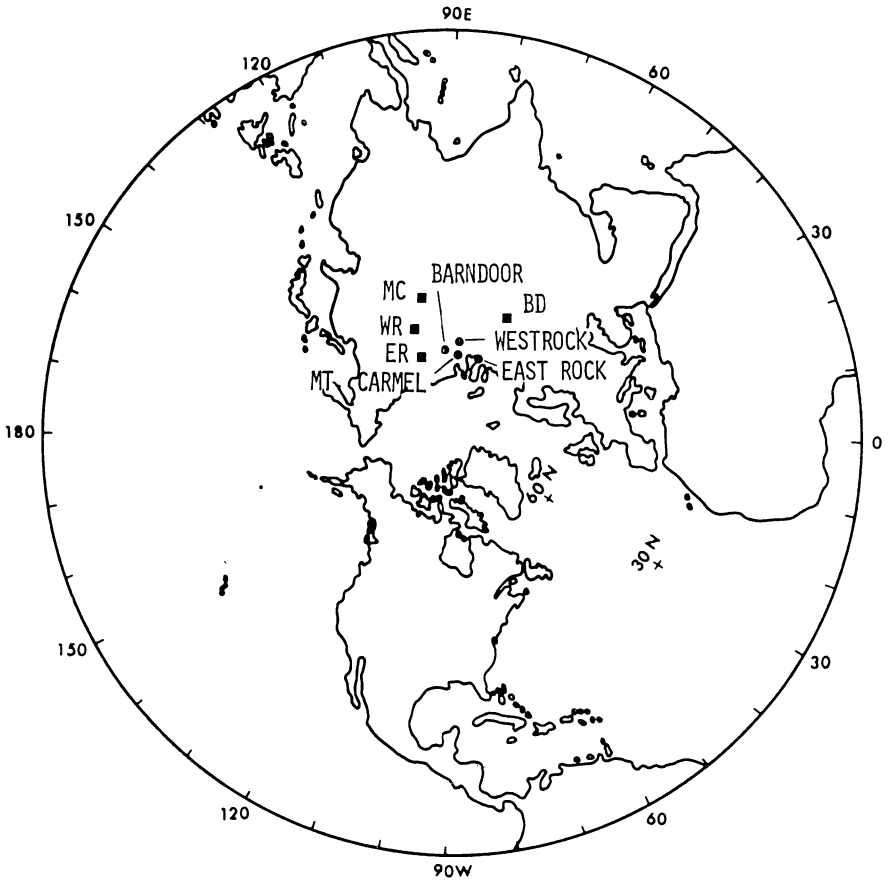


Fig. 6. Mean pole positions of the Barndoor, East Rock, Mt. Carmel, and West Rock intrusives. Solid squares represent mean pole positions with no structural correction for the East Rock, Mt. Carmel, and West Rock units and the structurally corrected results from the Barndoor unit. All data from table 2. Polar stereographic plot.

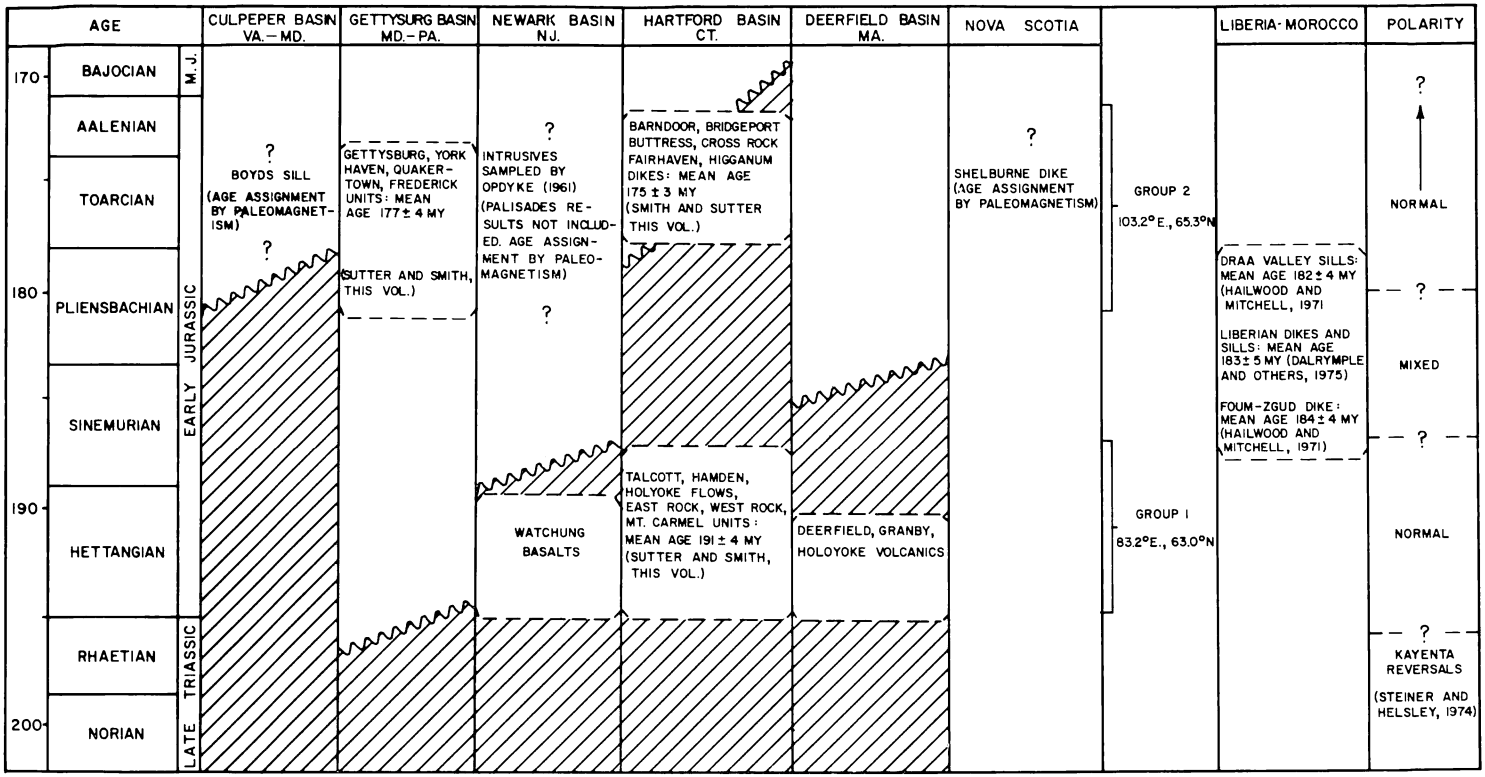


Fig. 7. Generalized correlation chart for the group 1 and group 2 results. Ages of sedimentary sections, bracketing ages of the basalt flows, and basin names are from Cornet (ms). Sedimentary sections are indicated by diagonal hatchuring. Details of unconformities in each basin have been omitted. Ages corresponding to the Triassic-Jurassic stages have been taken from Van Eysinga (1975). Age determinations from Newark trend rocks were obtained or calculated by Sutter and Smith (this volume). Ages and the mean age of the Liberian and Moroccan dikes and sills were obtained from Hailwood and Mitchell (1971) and Dalrymple, Grommé, and White (1975).

rocks. This age assignment is strongly supported by two $^{40}Ar/^{39}Ar$ isochron ages of 173 and 168 m.y. obtained from it (fig. 7) by Sutter and Smith (this issue). Addition of the Barndoor results to the mean pole position of the narrow dikes results in a new mean pole position located at 98.9°E, 68.4°N (table 2), representing a change of only 1.0°.

These results are surprising as the large size and conformable emplacement of the Barndoor intrusive suggest contemporaneity with the West Rock body (deBoer, 1968). Kroll (1976) has suggested a multiple intrusion history for the Barndoor body, but the paleomagnetic data are not able to resolve separate intrusive events.

The Connecticut paleomagnetic results indicate two igneous episodes separated by tilting of most of the Hartford Basin sediments. The two mean pole positions are separated by 8.4° and are statistically distinguishable. Recent palynologic, paleontologic, and radiometric age work (Cornet, ms; Olsen and Galton, 1977; Sutter and Smith, this issue) suggest that the igneous episodes are separated by at least 15 m.y. (fig. 7); the older event occurring during Hettangian-Sinemurian time (about 190 m.y.) and the younger event during Toarcian-Bajocian time (about 175 m.y.).

GROUP 1 AND 2 MEAN POLE POSITIONS AND APPARENT POLAR WANDER (APW)

Inspection of the paleomagnetic results from Newark trend rocks outside Connecticut reveals a tendency toward grouping into two clusters of VGPs with locations similar to the two sets of Connecticut results (app 1). The Watchung basalts of New Jersey and the Granby, Holyoke, and Deerfield volcanic units of Massachusetts (Opdyke, 1961; Bowker, ms; Irving and Banks, 1961) have paleomagnetic poles similar to those of the flows and East Rock, Mt. Carmel, and West Rock intrusives of Connecticut. Palynologic bracketing ages (Cornet, ms) show that they are of Hettangian-Sinemurian age, the same as the Connecticut units, suggesting that they form a recognizable time grouping (group 1) with them (fig. 8). Paleomagnetic results from the North Mountain basalts of the Fundy Group, Nova Scotia (Carmichael and Palmer, 1968) do not agree well with those above, but this may be due to secular variation effects arising

TABLE 3

Mean pole positions of Connecticut Newark trend igneous rocks

	N	k	α_{95}	E. Long	N. Lat
1. Mean pole position of the narrow dikes, including the Barndoor intrusive.	52	104	1.9°	98.9°	68.4°
2. Mean pole position of the East Rock, Mt. Carmel and Westrock intrusives.	25	107	2.8°	82.5°	63.1°
3. Mean pole position of the units in (2) and the Talcott, Holyoke and Hampden basalts.	59	68	2.3°	83.9°	62.6°

Mean pole positions obtained by averaging VGP's. N is number of site VGP's. Data sources are listed in the text and appendix 1.

from sampling through insufficiently thick sections (Carmichael and Palmer, 1968).

Most remaining paleomagnetic results from Newark trend igneous rocks of the north central Appalachians are similar to those from the Connecticut narrow dikes. These results (app 1) come from the Boyds sill and Frederick dikes of Maryland (Smith, 1976); the Gettysburg, York Haven, Quakertown, and Birdsboro sills of Pennsylvania and Maryland (Beck, 1972; Smith, 1976), various intrusives of New Jersey and Pennsylvania (Opdyke, 1961), and the Shelburne dike of Nova Scotia (Larochelle and Wanless, 1966).

Although palynologic bracketing ages cannot be determined for these intrusives, ages of host rocks, radiometric ages, and paleomagnetic directions suggest that they, along with the Connecticut narrow dikes, form a

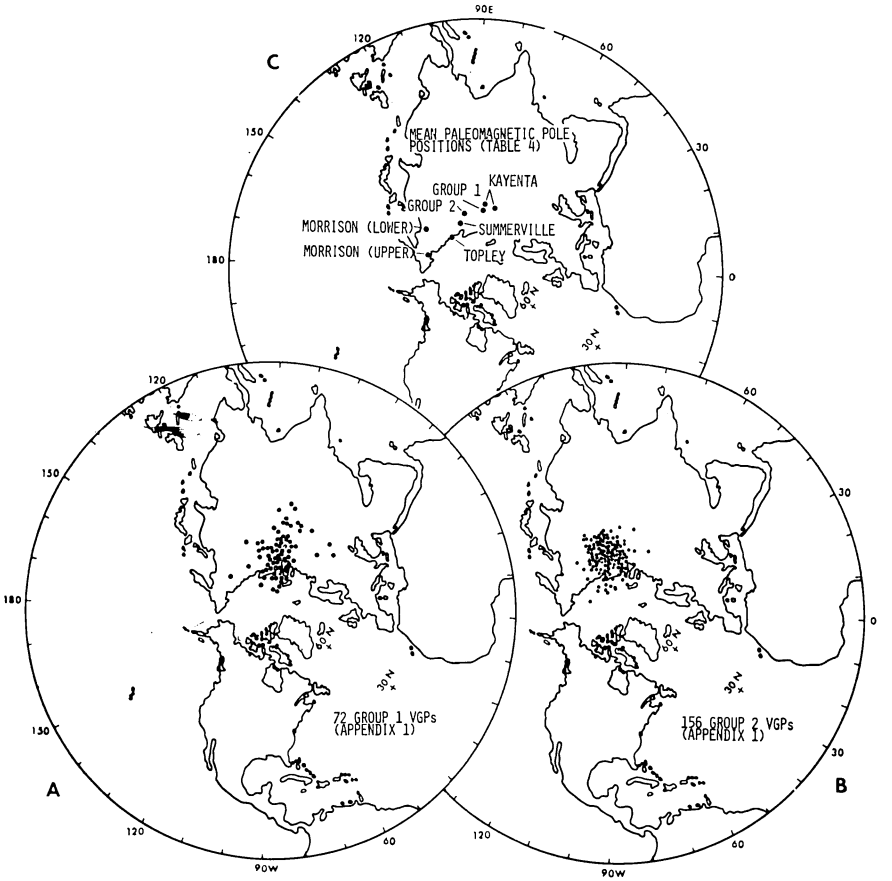


Fig. 8. A — Distribution of the 72 group 1 VGPs listed in appendix 1; B — Distribution of the 156 group 2 VGPs listed in appendix 1; C — Location of group 1 and group 2 mean pole positions and other Late Triassic-Jurassic North American mean paleomagnetic pole positions. Data concerning these pole positions are given in table 4. Polar stereographic plots.

second well defined time group (group 2) with a mean age of about 175 m.y. (fig. 7). The Boyds sill of Maryland was probably emplaced after rotation of the northern portion of the Culpeper basin sequence (Smith, 1976) indicating that it is probably not older than Pliensbachian-Toarcian (Early Middle Jurassic) age (Cornet, ms). Smith, Rose, and Lanning (1975) used geochemical data to show that the Frederick dikes, Gettysburg, York Haven, Quakertown, and Birdsboro sills of Maryland and Pennsylvania were intruded in a short time interval, probably within 10^6 yr. Five of six radiometric ages from the Frederick dikes and the Gettysburg sill (Sutter and Smith, this issue) cluster around 175 m.y. The remaining age is a discordant $^{40}Ar/^{39}Ar$ isochron age of 194 m.y. The five determinations suggest a Toarcian-Bajocian age for these units. Mean paleopole longitudes from these units (Beck, 1972; Smith, 1976) typically agree well with the Connecticut narrow dikes. However, the mean paleopole latitudes of the Gettysburg, York Haven, and Birdsboro units are about 4° to 6°S of those of the Connecticut dikes. This may be a result of slightly different intrusion (magnetization) time or somewhat excessive structural correction (assuming intrusion before any rotation of the basin rocks). Smith, Rose, and Lanning (1975) and Volk (1977) have demonstrated that these units and the Quakertown body are multiple intrusions consisting of two distinguishable diabase types; intrusion of one of these types following slight rotation of the basin rocks (5-8 degrees) would serve to reduce the anomalously low paleopole latitudes. This is suggested by the near parallelism of many dip directions in the Gettysburg Basin (the rocks generally dip 20°-30°NW) and the Early-Middle Jurassic paleofield declinations there (350°-10°).

Sparse paleomagnetic results from diabase intrusives of east Pennsylvania and New Jersey (Opdyke, 1961) and the Shelburne dike of Nova Scotia (Larochelle and Wanless, 1966) suggest they are equivalent to the previously mentioned Maryland and Pennsylvania intrusives and the Connecticut narrow dikes.

The mean pole positions of the group 1 and group 2 rocks (calculated by averaging VGPs) are located at 83.2°E, 63.0°N and 103.2°E, 65.3°N (table 4, fig. 8) and are separated by 9.0°. Both are characterized by good statistical quality and are distinct from each other, reinforcing the conclusion that the rocks contributing to them form two distinguishable time groups. The large number, geographical distribution, and age uncertainty of the units involved suggest that the two mean pole positions represent maxima of igneous activity spanning at least 5 m.y. and not "spikes" of short duration. Cornet (ms) indicates that the basalt flows of New Jersey, Connecticut, and Massachusetts, all contributing to the group 1 paleomagnetic pole position, may span up to 9 m.y.

Azimuthal and latitudinal symmetry tests using the χ^2 statistic (Watson and Irving, 1957) show that the group 1 VGPs form a Fisherian distribution (Fisher, 1953), whereas the distribution of group 2 VGPs is Fisherian only in terms of azimuthal symmetry. The latitudinal symmetry test shows that the VGPs are overly concentrated toward the mean.

The Fisherian angular standard deviation estimate, $S = 81/\sqrt{k}$, is 10.8° + 1.4°, −1.1° (95 percent limits following Cox, 1969) for the group 1 compilation and 8.4° + 0.7°, −0.6° for the group 2 compilation. Both values are less than expected for dispersion due to secular variation (Baag and Helsley, 1974). This is in agreement with CRM "smoothing" of many of the paleomagnetic results from the Newark trend igneous rocks, particularly so for the group 2 compilation. The somewhat higher dispersion associated with the group 1 pole position appears to result from inclusion of data from basalt flows, which probably became stably magnetized in a shorter amount of time than most of the intrusives and recorded more secular variation dispersion.

The two pole positions record about 9° of eastward apparent polar motion from Hettangian-Sinemurian to Toarcian-Bajocian time. Steiner and Helsley (1974), Johnson (1976), and Steiner (1975, 1978) show remarkably similar apparent polar motion from Upper Triassic to Callovian (Middle Jurassic) sedimentary rocks of the western United States (table 4). In addition, Symons (1973) and Steiner and Helsley (1975) show that continued eastward apparent polar motion on an irregular path between 60° and 70°N is characteristic of the Late Jurassic (table 4).

MAGNETIC POLARITY

The paleomagnetic results of this study and nearly all previous studies of the Newark trend igneous rocks (DuBois and others, 1957; Bowker, ms; Opdyke, 1961; Irving and Banks, 1961; LaRochelle and Wanless, 1966; deBoer, 1967, 1968; Carmichael and Palmer, 1968; Beck, 1972; Watts, ms; Smith, 1976; and Rigotti, 1976) are of normal polarity. The only reversed polarity results have been reported from isolated dikes

TABLE 4

North American Late Triassic(?)-Jurassic paleomagnetic data

Unit(s)	Age	N	k	α_{95}	dm	dp	E.Long	N.Lat.	Source
Kayenta fm., Utah	T_{R_U} (?)	105	32	2.5°	2.6°	1.4°	72.7°	61.7°	Steiner and Helsey (1974)
"	"	7	102	6.0°	-	-	82.5°	61.2°	Johnson (1976)
Group 1	T_{R_U} (?)-J (~190 m.y.)	72*	56	2.3°	-	-	83.2°	63.0°	This study
Group 2	J_l -J_m (~175 m.y.)	156*	92	1.4°	-	-	103.2°	65.3°	"
Summerville fm. Utah	J_m (Callovian)	15	71	4.6°	5.4°	3.2°	110.6°	67.5°	Steiner (1978)
Topley intrusives, Brit. Columbia	J_u	13*	12	9.1°	14.4°	11.4°	128.6°	72.0°	Symons (1973)
Morrison fm., Utah. (Lower).	J_u (Kimmeridgian)	32	23	5.4°	6.7°	4.1°	143.0°	61.4°	Steiner and Helsey (1975)
Morrison fm., Utah. (Upper).	"	68	35	3.0°	4.2°	3.0°	163.8°	67.8°	"

N is number of specimens except entries with asterisk. Asterisk denotes number of sites or VGP's. dm, dp are semi-axes of the 95 percent confidence oval surrounding the manpole position, applicable only for pole positions calculated directly from field directions.

in Rhode Island (Choiniere and Watkins, 1976) and Massachusetts (Smith, 1977). Hargraves (1975, personal commun.) encountered reversed magnetizations in the Norian-Sinemurian Brunswick Formation of New Jersey. Steiner and Helsley (1974) and Steiner (1978) present data showing that Late Triassic to early Middle Jurassic time was characterized by mixed polarity, but this is inconsistent with the nearly exclusively normal results from the Newark trend igneous rocks. This may be a result of (1) widespread remagnetization in normal polarity fields, (2) igneous activity only during periods of normal polarity, or (3) actual predominance of normal polarity during much of this time interval. The first choice may be possible but, if so, depends on predominantly normal polarity. There is no apparent reason for selective remagnetization only in normal fields. In addition, stable reversed results from the Medford diabase of Massachusetts (Smith, 1977), free of streaking and antipodal to normal results from the same unit, argue against thorough remagnetization of many of the Newark trend igneous rocks during periods of normal polarity. The second choice appears unlikely owing to the number of units that have been sampled, their geographic distribution, and the range of their ages. It receives support, though, from reversed results obtained by Dalrymple, Grommé, and White (1975) from Early Jurassic diabase dikes and sills of Liberia. These units have an age of 182 ± 4 m.y. (Hailwood and Mitchell, 1971; Dalrymple, Grommé and White, 1975), slightly older than the group 2 rocks (fig. 7). The third choice receives support simply from the overwhelming proportion of normal results.

These possibilities suggest the existence of two (at least 5 m.y. long) normal polarity intervals centered on 190 and 175 m.y. separated by a mixed polarity interval of unknown length (fig. 7). This conclusion should be regarded as speculative, though, and more sampling is needed to help resolve the situation.

PALEOMAGNETIC RECONSTRUCTION OF AFRICA AND NORTH AMERICA AND THE OPENING OF THE CENTRAL ATLANTIC OCEAN

Dalrymple, Grommé, and White (1975) showed that rotation of Africa against North America according to Bullard, Everett, and Smith (1975) and LePichon and Fox (1971) produced good convergence of a mean pole of Newark trend igneous rocks and a mean pole position of Late Triassic(?)-Early Jurassic intrusives of Liberia, Morocco, and Sierra Leone.

Their study was not able to employ Newark trend paleomagnetic data more recent than that by Beck (1972). In addition, Dalrymple, Grommé, and White (1975) included data from the Freetown complex of Sierra Leone (Briden, Heuthorn, and Rex, 1971) in their African pole position. The directions of this unit are considerably different from those of the Liberian and Moroccan units and suggest remagnetization in the present field.

Reassembly of Africa and North America according to either Bullard, Everett, and Smith (1965) or LePichon and Fox (1971) produces ex-

cellent convergence of the mean pole position of the Liberian and Moroccan results alone (table 5) with the group 2 pole position (fig. 9). The Liberian and Moroccan results used are given in appendix 2. The LePichon and Fox (1971) fit is visually superior although not statistically so, a conclusion consistent with Dalrymple, Grommé and White (1975). The angular separation of the two pole positions is only 2.5° for the LePichon and Fox (1971) arrangement and 3.5° for the Bullard, Everett, and Smith (1965) arrangement.

The rotated African pole position does not agree well with the group 1 pole position (fig. 9). The two pole positions are distinguishable for both reconstruction models. This may be an indication of slight continental motion during the time interval between emplacement of most of the two suites of rocks. This interval may have been up to 10 m.y. long.

Dalrymple, Grommé, and White (1975) suggested that very good agreement between North American and African pole positions following closure of the Central Atlantic Ocean is indicative of emplacement before separation and can be used to establish an earliest age of separation. Using this argument, the very good agreement between the group 2 and Liberian-Moroccan mean pole positions suggests that little or no separation had occurred by 175 m.y. ago. However, Steiner (1975) suggested that published rotation poles for the early opening of the ocean (Pitman and Talwani, 1972) are too close to the paleomagnetic pole to enable distinction between complete closure and slight opening. Rotation of the African results about Pitman and Talwani's (1972) pole for the earliest opening (180-155 m.y.) of the Central Atlantic Ocean, located at 73.3°E, 56.8°N, bear this out. This pole may not be accurately located, however, as rotation about it does not produce the Latest Triassic-Middle Jurassic portion of the APW path shown by the paleomagnetic results of this study and Steiner (1978) listed in tables 3 and 4. These results, representing about 190 to 160 m.y., lie on a small circle about a finite rotation pole located at 72.5°N, 86°E, only 16.5° distant from Pitman and Talwani's (1972) pole listed above. Unfortunately, this pole position also is too close to the paleomagnetic pole to enable paleomagnetic distinction between complete closure and slight opening of the ocean. (It should be

TABLE 5

Mean pole positions of Late Triassic(?)-Jurassic dikes and sills from Liberia and Morocco

Location of Africa	N	k	α-95	E. Long	N. Lat	Plong	Plat	Rotation angle
1.	46	44	3.2°	239.9°	66.3°	-	-	-
2.	"	"	"	97.9°	66.9°	348.0°	66.0	74.8°
3.	"	"	"	99.8°	68.5°	346.0°	67.6°	74.8°

1. Africa in present coordinates; 2. Africa rotated according to Le Pichon and Fox (1971); 3. Africa rotated according to Bullard, Everett, and Smith (1965).

Plong is east longitude of rotation pole, Plat is north latitude of rotation pole.

noted that the amount of rotation about this pole required to produce the paleomagnetic results, about 79°, is far too great to account for early opening of the ocean. This indicates that this rotation pole may be a finite representation of several instantaneous rotation poles.) Schouten (1977, personal commun.) suggests rotation of 61.1° about a finite rotation pole located at 66.2°E, 17.7°W for closure of the Central Atlantic Ocean to anomaly M-21 time or about 145 m.y. Rotation of the African results by this amount leaves them at 113.7°E, 73.7°N, which is distinguishable from the group 2 pole position. This indicates that the paleomagnetic data are able to resolve spreading of the ocean basin only after

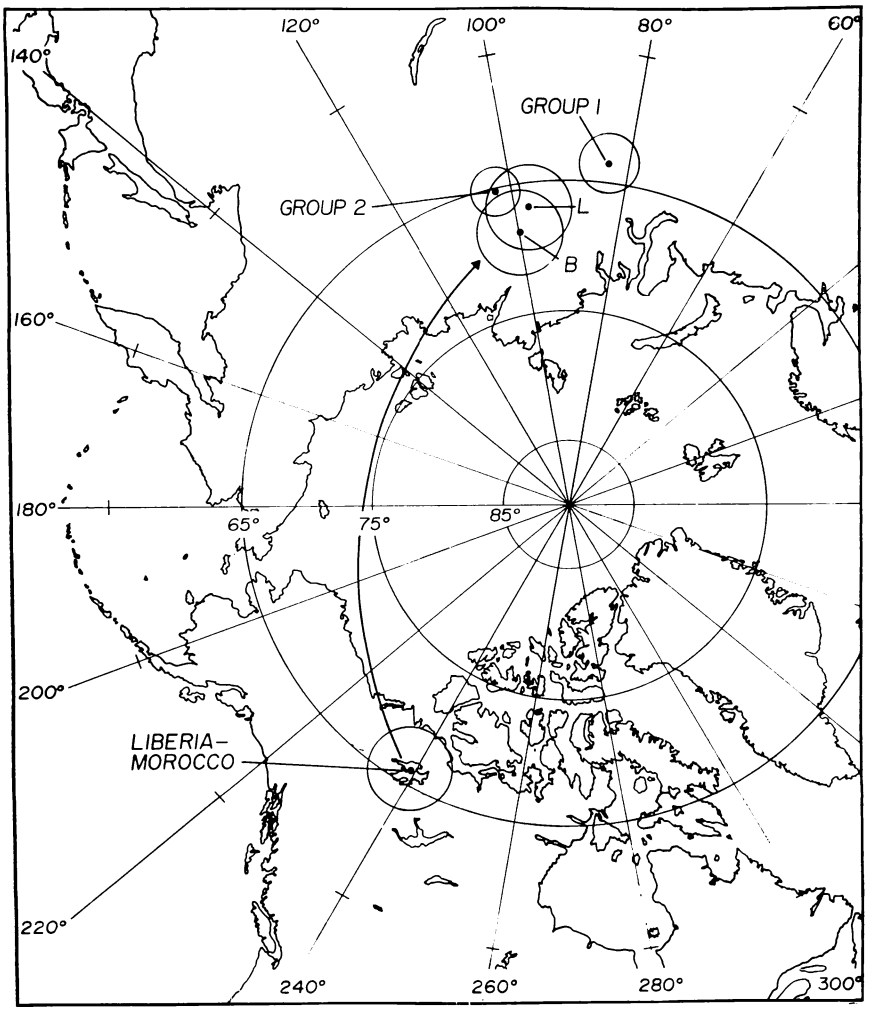


Fig. 9. Convergence of the mean pole positions of the Liberian and Moroccan intrusives with Group 1 and Group 2 mean pole positions according to the reconstructions of Bullard, Everett and Smith (1965) (B) and LePichon and Fox (1971) (L).

opening of about 1200 km, about one-fifth of its present width, consistent with Steiner's (1975) conclusion.

The poor resolving power of the paleomagnetic data reduces its role in determining the age of initial steady opening to a supportive nature. However, several lines of evidence, consistent with the paleomagnetic data, show that steady opening began between 170 and 175 m.y. ago. May (1971) considered that the numerous basic dikes in continental margins bordering the Central Atlantic Ocean describe an elliptical tensile stress field that immediately preceded steady opening of the ocean. The age data summarized in figure 7 show that many of these units are in the range of 170 to 175 m.y. Barrett and Keen (1976) estimated an age of 172 m.y. for the slope anomaly off Nova Scotia, based on spreading rates deduced from weak anomalies in the area (M-26, 27, and 28) dated at 154 to 164 m.y. Given (1977) summarized lithologic data from numerous boreholes off Nova Scotia and interpreted a major marine transgression in Early Middle Jurassic time, or about 170 to 175 m.y. ago, to be indicative of the onset of steady spreading. In addition, Cornet (ms) has shown that subsidence of the isolated basins of the Newark Supergroup had probably ceased by about 167 m.y. ago, which may have been a result of organization of the Central Atlantic spreading system by that time.

CONCLUSIONS

The main conclusions of this work are summarized below:

1. Paleomagnetic results from the East Rock, Mt. Carmel, and West Rock intrusives suggest contemporaneity with the Early Jurassic basalt flows of the Hartford Group of Connecticut, whereas those of the Barndoor intrusive suggest contemporaneity with the Early-early Middle Jurassic narrow dikes of Connecticut. These results indicate two rather well defined periods of igneous activity accompanying Newark style sedimentation and tectonism in Connecticut, one preceding and the other following tilting of the basin sediments.

2. Many of the Newark trend igneous rocks elsewhere in the north central Appalachians have paleomagnetic pole positions and associated ages corresponding to those of the Connecticut units, suggesting that the igneous maxima or pulses recorded there were of regional extent. Combining the paleomagnetic results of these units with those from Connecticut produces a mean Early Jurassic pole position (group 1, corresponding to about 190 m.y.) located at 83.2°E, 63.0°N and a mean early Jurassic pole position (group 2, corresponding to about 175 m.y.) located at 103.2°E, 65.3°N.

3. The two mean pole positions record slow eastward apparent polar motion during the Late Triassic?-early Middle Jurassic, in very good agreement with results from paleomagnetic studies on rocks of similar age in the western United States (Steiner and Helsley, 1974; Johnson, 1976; and Steiner, 1975, 1978).

4. The Newark trend igneous rocks sampled thus far are nearly exclusively of normal polarity. These results and mixed results from Early

Jurassic (about 180 m.y.) intrusives of Liberia show the existence of long (probably 5 m.y. or more) normal intervals at about 190 and 175 m.y. separated by an interval of mixed polarities of unknown but possibly short duration.

5. Rotation of Africa against North America produces excellent convergence of a mean pole position of Early Jurassic intrusives of Liberia and Morocco with the group 2 mean pole position, but apparent close proximity of rotation poles for the early opening of the Central Atlantic Ocean with the magnetic pole does not permit paleomagnetic distinction between complete closure and slight opening of the ocean. However, independent offshore borehole and sea floor magnetic anomaly data (Given, 1977; Barrett and Keen, 1976) and structural interpretation (May, 1971; Cornet, ms) show that the onset of steady spreading in the Central Atlantic Ocean probably began between 170 and 175 m.y. ago, an age interval with which the paleomagnetic results are highly consistent.

ACKNOWLEDGMENTS

We are grateful to B. Cornet for his information on the palynostratigraphy, structure, and age of the Newark Supergroup. We also thank M. Kopacz, B. McFadden, M. Rhodes, and M. Steiner for reviews and comments which directly and indirectly led to improvements in the manuscript.

APPENDIX 1

VGPs used in calculating the mean pole positions of the group 1 and 2 rocks

Group 1

Pole position E. Long	N. Lat	unit	N	α-95	Reference
82.3	49.9	Talcott basalt, Connecticut	—	—	deBoer (1968)*
83.5	50.7	"	—	—	"
74.2	49.5	"	—	—	"
81.5	51.3	"	—	—	"
79.1	52.4	"	—	—	"
71.4	50.9	"	—	—	"
71.0	51.4	"	—	—	"
82.0	56.5	"	—	—	"
99.8	59.2	Holyoke basalt, Connecticut	—	—	"
86.6	58.4	"	—	—	"
83.0	57.8	"	—	—	"
87.1	55.8	"	—	—	"
83.0	60.9	"	—	—	"
78.9	60.7	"	—	—	"
89.9	63.7	"	—	—	"
80.8	63.7	"	—	—	"
95.1	67.4	"	—	—	"
96.1	64.8	Hampden basalt, Connecticut	—	—	"
88.0	66.8	"	—	—	"
120.2	68.1	"	—	—	"
143.3	65.5	"	—	—	"
93.4	70.8	"	—	—	"

Pole position E. Long	N. Lat	unit	N	α-95	Reference
83.9	71.5	"	—	—	"
88.1	73.5	"	—	—	"
85.1	73.2	"	—	—	"
70.3	72.1	"	—	—	"
72.8	74.4	"	—	—	"
115.6	77.4	"	—	—	"
72.5	76.3	"	—	—	"
70.6	77.1	"	—	—	"
88.1	79.7	"	—	—	"
97.2	68.5	Watchung basalts, New Jersey	12	4.4	Opdyke (1961)*
98.8	61.9	"	8	7.0	"
69.5	58.8	"	10	3.5	"
42.6	52.9	"	8	2.6	"
81.7	65.2	"	12	2.6	"
74.8	65.2	Granby tuff, Massachusetts	6	5.1	Irving and Banks (1961)*
113.3	58.3	"	6	3.6	"
105.7	65.7	Holyoke basalt, Massachusetts	6	5.7	"
78.4	45.3	"	8	5.3	"
87.1	46.2	"	6	11.9	"
90.0	54.0	"	8	11.0	DuBois and others (1957)
74.7	80.0	West Rock intrusive, Connecticut	9	4.5	This study
89.7	66.7	"	13	2.3	"
81.4	69.8	"	8	1.9	"
106.8	71.8	"	7	7.7	"
84.4	65.4	"	10	4.2	"
83.8	64.5	"	9	3.9	"
103.1	62.9	"	8	2.9	"
52.1	57.6	"	6	7.8	"
84.4	59.4	"	3	12.9	deBoer (1968)*
91.8	63.1	"	4	4.8	"
87.2	65.5	"	4	8.1	"
87.6	63.3	Mt. Carmel intrusive, Connecticut	14	3.6	This study
77.5	71.1	"	10	2.2	"
72.7	59.6	"	9	4.6	"
101.7	59.5	"	6	8.0	DeBoer (1968)*
79.1	56.7	"	6	8.4	"
86.6	59.4	"	7	8.1	"
95.1	61.8	"	7	6.4	"
100.7	70.9	"	6	7.9	"
70.1	69.2	East Rock intrusive, Connecticut	9	3.1	This study
66.0	65.6	"	9	3.4	"
71.0	68.7	"	9	3.2	"
74.9	66.7	"	8	4.7	"
80.4	71.9	"	12	3.0	"
64.1	61.0	"	14	2.5	"
80.0	44.0	Deerfield basalt, Massachusetts	16	5.0	Bowker (ms)**
50.0	50.0	"	13	11.0	"
69.0	72.0	Hampden basalt, Connecticut	12	6.0	"
80.0	58.0	Talcott basalt, Connecticut	7	9.0	"
75.0	63.0	Holyoke basalt, Connecticut	5	5.0	"

*Calculated from data given in the original reference. **Recalculated by deBoer (1968).

Group 2

Pole position E. Long	N. Lat	unit	N	α-95	Reference
105.0	55.6	Diabase intrusives, New Jersey-Pennsylvania	13	8.5	Opdyke (1961)*
94.1	59.6	"	12	6.7	"
149.1	60.4	"	12	7.3	"
116.4	62.7	"	12	8.1	"
109.8	66.7	"	8	5.3	"
89.0	66.2	Barndoor intrusive, Connecticut	10	4.5	This study
101.5	63.3	"	11	4.6	"
76.6	62.3	"	11	2.2	"
85.3	58.0	"	6	7.4	"
115.5	64.7	"	8	6.3	"
93.9	65.4	"	9	9.7	"
110.9	63.1	"	8	1.9	"
81.6	67.3	"	8	7.4	"
104.9	71.2	"	5	6.7	"
116.8	63.6	"	9	5.9	"
89.2	62.5	"	7	3.2	"
111.9	63.6	Bridgeport dike, Connecticut	5	4.6	Smith (1976)
104.5	63.3	"	6	5.0	"
87.2	72.3	"	7	3.3	"
87.7	68.7	"	15	6.2	"
80.3	69.8	"	6	9.2	"
100.5	69.1	"	10	3.6	"
94.2	62.3	"	9	2.6	"
91.2	64.1	"	9	3.4	"
105.6	71.5	Buttress dike, Connecticut	13	4.1	"
72.1	78.5	"	15	1.3	"
107.7	71.1	"	14	3.3	"
95.7	78.8	"	9	5.6	"
82.3	75.2	"	9	6.1	"
99.7	83.5	"	9	4.4	"
89.1	70.0	"	7	2.9	"
106.2	67.9	"	8	2.8	"
70.5	77.0	"	7	3.7	"
95.1	69.4	Cross Rock dike, Connecticut	8	3.0	"
98.1	64.5	"	10	6.9	"
134.3	77.6	"	6	4.3	"
106.3	78.2	"	7	3.6	"
129.3	75.6	"	6	4.4	"
130.8	65.9	"	11	7.1	"
100.7	65.4	Fairhaven dikes, Connecticut	10	2.3	"
118.8	58.3	"	7	7.2	"
120.5	79.8	"	9	3.9	"
117.6	73.6	"	9	4.0	"
63.8	60.3	"	9	3.3	"
87.5	62.5	"	11	3.1	"
110.9	67.5	"	11	5.0	"
92.4	67.3	Higganum dike, Connecticut	7	9.8	"
84.9	64.4	"	11	4.7	"
94.7	61.7	"	7	4.0	"
88.5	60.8	"	8	2.3	"
108.7	64.7	"	11	2.6	"
115.9	66.9	"	10	2.6	"
122.7	66.4	"	9	4.0	"
115.3	65.3	"	10	3.2	"
89.3	63.7	"	12	7.7	"
103.7	70.3	"	21	4.1	"
105.3	67.5	"	9	1.7	"
119.7	63.7	Boyds sill, Maryland	5	1.2	"

Pole position E. Long	N. Lat	unit	N	α-95	Reference
107.2	64.9	"	5	3.8	"
104.2	66.9	"	5	3.0	"
110.3	74.4	"	8	3.8	"
93.4	68.1	"	7	2.7	"
103.8	66.5	"	8	3.2	"
108.1	65.1	"	6	3.3	"
91.7	61.0	Frederick dikes, Maryland	17	2.5	"
102.8	65.0	"	20	2.2	"
111.0	71.5	"	22	2.0	"
103.9	68.0	"	17	1.9	"
102.7	67.8	"	14	1.8	"
116.6	71.5	"	12	5.8	"
88.2	72.3	"	8	8.6	"
115.3	70.2	"	16	4.5	"
113.6	71.8	"	12	5.3	"
97.4	64.2	"	14	1.3	"
91.6	59.1	Gettysburg intrusive, Maryland-Pennsylvania	8	2.4	"
98.9	59.7	"	7	2.3	"
96.9	59.9	"	8	1.7	"
115.3	55.1	"	5	4.2	Beck (1972)*
161.2	76.9	"	4	2.9	"
84.6	65.7	"	5	9.9	"
90.6	61.4	"	4	3.7	"
106.8	58.4	"	5	2.1	"
100.4	66.7	"	5	12.7	"
99.8	61.9	"	5	6.0	"
115.5	62.1	York Haven intrusive, Pennsylvania	4	7.4	"
108.2	62.5	"	4	9.0	"
107.0	61.1	"	4	9.6	"
115.2	58.0	"	4	12.0	"
104.0	60.3	"	4	9.9	"
118.8	59.9	"	4	12.9	"
115.7	59.6	"	5	9.9	"
110.9	65.1	"	5	10.0	"
99.7	63.3	"	5	7.2	"
91.4	67.6	"	5	13.3	"
106.4	67.0	"	5	5.8	"
115.7	59.6	"	5	10.7	"
111.6	58.5	"	5	12.6	"
112.6	59.0	"	4	3.8	"
103.1	58.2	"	5	14.8	"
107.8	58.9	"	5	2.9	"
105.9	58.1	"	5	4.6	"
108.7	58.4	"	5	7.6	"
106.2	63.9	"	5	5.9	"
107.4	64.3	"	4	6.7	"
105.1	63.7	"	4	11.3	"
108.4	63.9	"	5	6.8	"
89.6	64.0	"	5	2.2	"
89.8	53.0	"	5	2.4	"
104.0	59.5	"	4	3.3	"
97.2	59.6	"	5	7.7	"
111.1	61.0	"	5	2.3	"
94.8	57.1	"	4	14.2	"
99.1	59.9	"	5	4.4	"
102.2	56.3	"	5	7.2	"
117.8	54.5	"	4	3.9	"

Pole position E. Long	N. Lat	unit	N	α-95	Reference
102.1	58.1	"	5	2.8	"
109.1	61.1	"	5	2.7	"
96.7	66.9	"	5	4.9	"
97.5	63.3	"	5	3.1	"
99.3	66.0	"	5	2.3	"
101.2	58.4	"	5	2.8	"
115.1	54.0	Birdsboro intrusive, Pennsylvania	5	11.6	"
102.3	54.8	"	5	9.4	"
101.7	65.5	"	5	8.8	"
115.0	60.5	"	5	13.3	"
102.2	56.1	"	5	10.3	"
105.8	56.3	"	5	14.2	"
105.0	59.2	"	5	11.0	"
111.1	55.9	"	5	5.7	"
105.0	58.9	"	5	13.6	"
132.7	62.0	"	5	11.0	"
119.0	67.4	"	5	6.0	"
116.1	74.7	"	5	9.1	"
135.7	70.4	"	5	4.6	"
97.4	58.3	"	4	7.0	"
84.7	58.7	"	4	4.8	"
79.4	53.2	"	4	11.9	"
107.9	68.6	"	4	10.0	"
123.2	68.0	"	4	9.2	"
108.5	64.5	"	5	9.5	"
96.3	54.0	"	4	11.0	"
108.5	56.2	"	5	4.2	"
92.0	57.3	"	5	5.4	"
80.4	62.6	"	5	6.0	"
112.1	65.3	"	5	8.8	"
93.5	68.7	"	5	2.7	"
120.6	67.9	Quakertown intrusive, Pennsylvania	5	6.8	"
107.0	72.0	"	5	8.6	"
101.4	68.6	"	5	6.5	"
119.3	78.3	"	5	6.7	"
89.1	62.4	"	5	5.0	"
100.0	63.2	"	5	2.0	"
107.4	64.8	"	5	7.4	"
101.9	62.8	"	5	6.1	"
83.3	64.8	"	5	6.0	"
98.0	69.0	Shelburne dike, Nova Scotia (mean of two sites)	11	4.0	Larochelle and Wanless (1966)

*Calculated from data given in the original reference.

APPENDIX 2

VGPs used in calculating the mean pole position of the Early Jurassic Liberian and Moroccan diabase dikes and sills

Pole position E. Long	Pole position N. Lat	unit	N	α-95	Reference
220.6	75.1*	dikes emplaced in Liberian age province, Liberia	10	2.7	Dalrymple, Grommé and White (1975)
252.8	64.0*	"	10	2.8	"
259.9	62.6*	"	10	2.2	"
227.7	53.3*	"	10	3.3	"
251.7	66.7*	"	9	2.6	"
264.2	60.2*	"	10	1.8	"
264.7	70.0*	"	8	4.5	"
282.7	68.6*	"	10	2.9	"
208.7	84.4*	"	9	4.6	"
238.5	48.5*	"	9	4.8	"
204.0	68.4*	"	10	3.4	"
249.6	69.8*	"	9	4.4	"
257.4	57.2*	dikes and sills emplaced in Paynesville sandstone and Pan African age province, Liberia (normal polarity)	10	8.6	"
220.8	74.6*	"	8	5.9	"
165.6	76.2*	"	10	2.3	"
200.7	75.5*	"	10	4.7	"
205.1	78.1*	"	10	3.0	"
197.8	68.0*	"	10	1.6	"
231.8	68.5*	"	9	4.3	"
269.9	52.1*	dikes and sills emplaced in Paynesville sandstone and Pan-African age rocks, Liberia (reversed polarity)	10	2.1	"
206.7	56.7*	"	16	3.1	"
250.5	64.8*	"	10	2.7	"
252.0	67.9*	"	10	2.3	"
257.3	45.7*	"	10	3.1	"
257.3	75.0*	"	10	3.0	"
215.0	68.0*	Draa Valley sills, Morocco	5	3.0	Hailwood and Mitchell (1971)
215.0	64.5	"	5	6.0	"
225.5	64.5	"	5	3.0	"
225.0	65.5	"	4	9.0	"
203.5	71.5	"	5	10.0	"
200.0	64.5	"	6	12.0	"
228.0	57.5	"	6	5.0	"
235.5	59.0	"	6	3.0	"
241.5	58.5	"	6	7.0	"
234.5	66.0	"	6	3.0	"
239.5	64.0	"	6	5.0	"
260.5	70.5	"	5	5.0	"
241.0	63.0	"	6	4.0	"
222.5	71.5	"	6	2.0	"
243.0	61.5	"	6	6.0	"
248.0	63.5	"	6	6.0	"
251.0	58.5	Foum-Zguid dikes, Morocco	5	8.0	"
252.5	58.5	"	5	3.0	"
263.5	55.0	"	6	8.0	"
265.5	59.5	"	3	11.0	"
261.0	58.0	"	4	12.0	"

*Calculated from data given in the original reference.

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